

HERO scorecard (vertical) — value ± BCa 95% CI · colour = per-metric min-max (green = best) · * = Δ-vs-frozen CI excludes 0
9 metrics × 8 encoders · WINNER col + BLUE BOX = surgery-vs-pretrain champion duel (tie if within ε) · 10 K bootstrap

	frozen	pre-2X	pre-enc	pre-hd	s3DI-enc	s3DI-hd	sNoDI-enc	sNoDI-hd	WINNER
action_top1 ↑ better	44.384 ±2.274	53.151 ±2.274*	51.671 ±2.274*	42.849 ±2.280	50.301 ±2.274*	51.233 ±2.301*	51.781 ±2.274*	51.726 ±2.301*	pretrain_2X encoder
motion_cos ↑ better	0.009 ±0.001	0.169 ±0.004*	0.159 ±0.004*	0.013 ±0.001*	0.163 ±0.004*	0.167 ±0.005*	0.168 ±0.005*	0.179 ±0.005*	surgery noDI head
taxonomy_f1 ↑ better	0.793 ±0.016	0.786 ±0.017	0.791 ±0.017	0.793 ±0.017	0.793 ±0.017	0.789 ±0.017	0.790 ±0.017	0.788 ±0.017	tie
future_mse ↓ better	0.557 ±0.001	0.546 ±0.001*	0.540 ±0.001*	0.558 ±0.001*	0.515 ±0.001*	0.540 ±0.001*	0.515 ±0.001*	0.540 ±0.001*	surgery 3stage_DI encoder
rollout ↓ better	0.009 ±0.000	0.009 ±0.000*	0.008 ±0.000*	0.009 ±0.000	0.009 ±0.000*	0.008 ±0.000*	0.009 ±0.000*	0.008 ±0.000*	tie
causal ↓ better	0.583 ±0.001	0.571 ±0.001*	0.565 ±0.001*	0.583 ±0.001	0.547 ±0.002*	0.565 ±0.001*	0.545 ±0.002*	0.565 ±0.001*	surgery noDI encoder
tdist ↓ better	0.007 ±0.000	0.008 ±0.000*	0.007 ±0.000	0.007 ±0.000	0.008 ±0.000*	0.007 ±0.000	0.008 ±0.000*	0.007 ±0.000	tie
teacher_free ↓ better	0.046 ±0.000	0.044 ±0.001*	0.043 ±0.001*	0.046 ±0.000	0.048 ±0.001*	0.043 ±0.001*	0.047 ±0.001*	0.043 ±0.001*	tie
maskratio ↓ better	0.072 ±0.001	0.071 ±0.001*	0.068 ±0.001*	0.072 ±0.001	0.059 ±0.001*	0.068 ±0.001*	0.062 ±0.001*	0.068 ±0.001*	surgery 3stage_DI encoder